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ORIGINAL PAPER

PathoSense: a rapid electroanalytical device platform for screening *Salmonella* in water samples

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Abstract
Salmonella contamination is a major global health challenge, causing significant foodborne illness. However, current detection methods face limitations in sensitivity and time, which mostly rely on the culture-based detection techniques. Hence, there is an immediate and critical need to enhance early detection, reduce the incidence and impact of *Salmonella* contamination resulting in outbreaks. In this work, we demonstrate a portable non-faradaic, electrochemical sensing platform capable of detecting *Salmonella* in potable water with an assay turnaround time of ~ 9 min. We evaluated the effectiveness of this sensing platform by studying two sensor configurations: one utilizing pure gold (Au) and the other incorporating a semiconductor namely a zinc oxide thin film coated on the surface of the gold (Au/ZnO). The inclusion of zinc oxide was intended to enhance the sensing capabilities of the system. Through comprehensive experimentation and analysis, the LoD (limit of detection) values for the Au sensor and Au/ZnO sensor were 0.9 and 0.6 CFU/mL, respectively. In addition to sensitivity, we examined the sensing platform's precision and reproducibility. Both the Au sensor and Au/ZnO sensor exhibited remarkable consistency, with inter-study percentage coefficient of variation (%CV) and intra-study %CV consistently below 10%. The proposed sensing platform exhibits high sensitivity in detecting low concentrations of *Salmonella* in potable water. Its successful development demonstrates its potential as a rapid and on-site detection tool, offering portability and ease of use. This research opens new avenues for electrochemical-based sensors in food safety and public health, mitigating *Salmonella* outbreaks and improving water quality monitoring.

Keywords Electrochemical sensors · Label-free detection · *Salmonella typhimurium* · Electrochemical impedance spectroscopy (EIS) · Potable water analysis

Introduction

Foodborne illnesses acquired through contaminated food and water are a biggest public health concern [1]. According to the World Health Organization (WHO), these illnesses cause approximately 10% of the world population to get sick and about 33 million people die each year. These illnesses tend to affect children more adversely. Out of these foodborne illnesses, diarrheal-causing diseases are the most common type, affecting about 550 million people (220 million of which are children under the age of 5) annually [1]. *Escherichia coli*, *Shigella*, *Listeria monocytogenes*, *Vibrio cholera*, and *Salmonella* are some of the most common types of bacteria that can cause outbreaks [1, 2]. Among these, *Salmonella* is one of the most common foodborne pathogens [3, 4].

The *Salmonella* bacterium is a gram-negative, rod-shaped bacterium that falls under the Enterobacteriaceae family. *Salmonella* is an especially dangerous bacterium because it is facultatively anerobic, meaning it can survive with oxygen (via aerobic respiration) and without oxygen (via fermentation). Out of 2500 different serotypes of this bacterium, *Salmonella enterica serovar typhimurium* (*S. typhimurium* or *Salmonella*) is the most prevalent pathogenic serotype found in humans [5]. Ingesting these bacteria can lead to salmonellosis with symptoms including fever, abdominal pain, nausea, and diarrhea [6]. These symptoms can persist anywhere from 2 to 7 days [7] [1]. Foodborne illnesses are

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difficult to isolate and pinpoint when bacteria are present in food and water. Due to the difficulty of tracing a wide variety of food groups consumed, identifying the cause of the outbreak is nearly impossible until the outbreak is at an outbreak level [8, 9]. Consequently, *Salmonella*-testing before food and water are distributed to consumers is the most effective way to prevent outbreaks. With this in mind, food and water regulatory committees maintain strict standards for the amount of bacteria in edibles. According to the US FDA, the ingredient should test negative for *Salmonella*. Similarly, the EU has established guidelines regarding food safety limits for *Salmonella* contamination, specifying that no *Salmonella* should be present in 25 g of any food, including water, meat, animal products, and seafood [2, 8].

The current methods for detecting *Salmonella* bacteria, which include culturing and plating, are precise but require multiple steps that can take up to 3–5 days [4, 9]. In order to reduce this time to less than 2 days, PCR, multiplex PCR, and quantitative PCR can be utilized to detect bacteria [10, 11]. However, the major issue with these methods is that they frequently yield false positives due to the complexity to differentiate between dead and live bacteria [12]. Using immunochromatographic strips and ELISA (enzyme-linked immunosorbent assay) can yield quick bacterial detection results but at the cost of not being sensitive enough to meet regulatory standards [13, 14]. To enhance these methods, they need to be coupled with a bacterial enrichment step, involving culturing process followed by an elective process of concentrating the bacteria via immunomagnetic separation or filtration [15]. Most of these methods have a limit of detection between 1 and 10 CFU/25 g [16]. Despite the high accuracy and precision of these methods, there are many drawbacks, including requiring multiple time-consuming steps, the necessity of limited shelf-life products, and the necessity for a highly skilled person [17]. Impedance-based electrochemical methods aim to reduce the drawbacks of the previously mentioned methods, and they are widely used for bacterial detection [18]. These methods employ electrochemical sensors that are extremely sensitive and selective, quick, user-friendly, and inexpensive [19–24]. Due to its simplistic nature among available sensors, label-free electrochemical impedance spectroscopy (EIS) is an extraordinary method to investigate molecular interactions [25–27]. A wide range of metal composites with a faradaic electrochemical approach has been found to enhance sensitivity and selectivity [28–30]. Supplementary Table S1 presents a comparison of different immunosensors, revealing that label-free variants lack the sensitivity for direct *Salmonella* detection. For this reason, the additional step of pre-concentration (via filtration or immunomagnetic concentration) or pre-enrichment process is necessary.

In this study, we demonstrate the utilization of a non-faradaic impedance spectroscopic approach towards developing a simple and easy immunoassay to detect the presence of *Salmonella* in potable water using a label-free approach. We have compared the non-faradaic electrochemical activity for metal (gold) and metal/semiconductor (gold/zinc oxide) surfaces for a single-capture immunoassay system whereby capture antibodies are immobilized on the sensor electrode and a single-capture system result in the direct label-free detection of the target. These electrode surfaces were functionalized with a thiol-based crosslinker allowing for the antibody to bind to the crosslinker. This allows the sensor to be able to detect the presence of *Salmonella* in spiked water samples. The sensor response was measured using EIS [24, 31, 32] across a broad frequency range, which provides insights into the electrochemical binding associated with the single-capture immunoassay. We conducted a comparative analysis of results obtained from two sensor types, namely Au and Au/ZnO. This comparative study yields valuable insights that can propel the advancement and refinement of more sophisticated electrochemical systems suitable for real-world applications. While point-of-care testing is prevalent in disease diagnosis, having the technological capability to extend this approach to test for disease-causing pathogens is equally crucial. In this groundbreaking work, we introduce a non-faradaic electrochemical platform utilizing label-free impedance spectroscopy for the rapid quantification of *Salmonella* through the detection of the coat protein. This innovative approach, demonstrated here for the first time, showcases the feasibility of designing ultra-sensitive platforms, as indicated by the limit of detection (LoD), without the need for signal amplification using an electrochemical tag or label.

Materials and methods

Materials and reagents

The polyclonal *Salmonella* antibody, sourced from Invitrogen (USA), was utilized at a concentration of 10 µg/mL. *Salmonella typhimurium* cells, heat-killed, were procured from Cedarlane (native antigen) and were applied at varying concentrations ranging from 10^1 to 10^6 colony-forming units per milliliter (CFU/mL). The crosslinker Dithiobis (sulfosuccinimidyl propionate) (DTSSP) was employed at a concentration of 10 mM and the phosphate buffer saline (PBS) with a pH of 7.4 was both obtained from Thermofisher Scientific Inc. (Waltham, MA, USA). The samples were partitioned into multiple parts and preserved at − 20 °C until they were used. To conduct the experiment, the smaller samples were brought to room temperature and centrifuged. Analytical-grade chemicals were used in this study. Therefore, no further purification process.

Sensor fabrication

The construction of the sensor involves the utilization of a portable printed circuit board (PCB) platform, implementing a dual-sensor system. In the first sensor, the working electrode comprises gold (Au) surfaces, while the second sensor features a modified gold electrode with a coating of zinc oxide (ZnO) nanomaterials. The copper-clad boards undergo a process of electroless nickel immersion gold (ENIG) application for both the laboratory-designed Au sensor and the Au/ZnO-sensor. This sensor is meticulously fabricated on a PCB substrate. An optical micrograph of the surface was captured with a HIROX microscope (series RH 2000) to show the Au and Au/ZnO sensing platform as shown in Fig. 1C.

ATR-IR spectroscopy setup

The acquisition of infrared spectra (IR) in attenuated total reflectance mode was performed using the Nicolet iS-50 FTIR instrument, manufactured by Thermo Scientific Inc. The detector employed deuterated triglycine sulfate, and the window material utilized was KBr. A germanium crystal was employed for spectrum collection, offering a resolution of 4 cm^{-1} within the wavelength range spanning from 4000 to 400 cm^{-1}, encompassing 256 scans. Following the same deposition parameters as the sensor substrate, Au and Au/ZnO were deposited on glass substrates. Just like the sensor immunoassay detailed in the "Electrode modification" section, DTSSP was used to functionalize and immobilize the Au and Au/ZnO electrodes. *Salmonella* antibodies were separately incubated on the electrode surfaces in 100% PBS.

Zeta potential measurements

Malvern Zetasizer NanoZS (Malvern Instruments, UK) was utilized to examine the zeta potential (ζ) of *Salmonella* bacteria and *Salmonella* antibodies at various concentration dilutions in potable water. The zeta potential was recorded for control (ZD), 10^1, 10^2, 10^3, 10^4, 10^5, and 10^6 CFU/mL doses of *Salmonella* in potable water, followed by antibody (10 µg/mL) binding with different concentrations of *Salmonella*. The Smoluschowski approximation method was applied to calculate the zeta potential values based on the measured electrophoretic mobility values.

Surface plasmon resonance (SPR) studies

For detailed SPR characterization studies, the OpenSPR instrument by Nicoya in Canada was utilized. Protein-A immobilization kit and carboxyl chip were used to thoroughly characterize the *Salmonella* antibody. The procedure encompassed meticulous surface cleaning with 10 mM HCl (pH 2 – 3), succeeded by EDC-NHS immobilization.

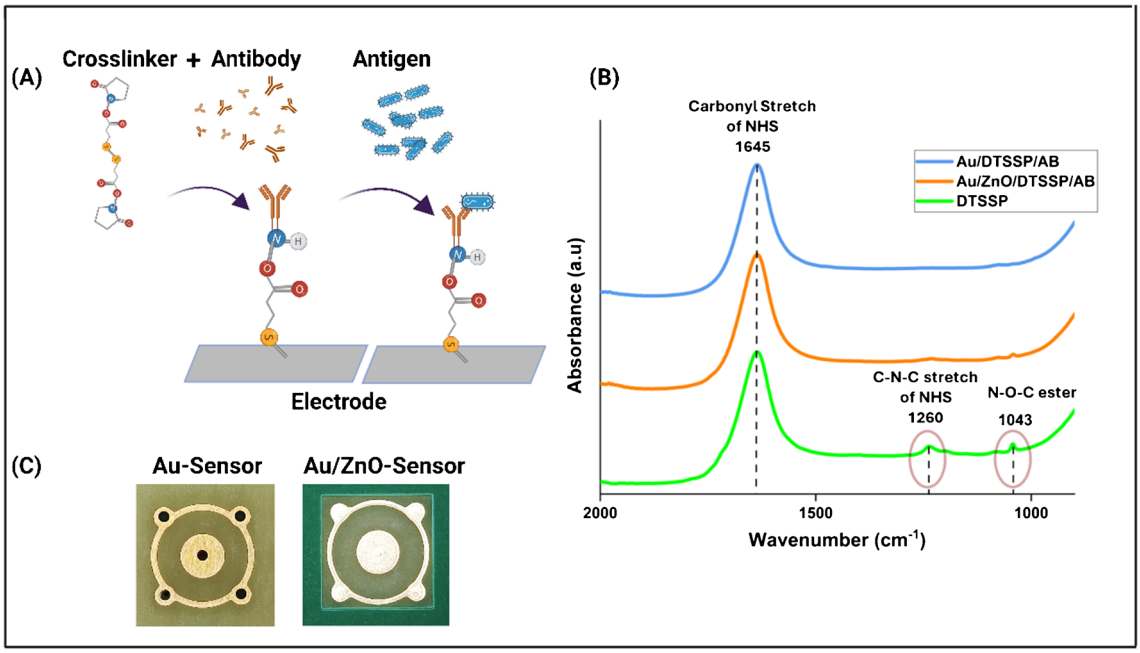

Fig. 1 **A** The FTIR spectra of the gold electrode subsequent to the immobilization of the thiol-based crosslinker (green spectrum), *Salmonella* antibody immobilized on Au/ZnO surface (orange spectrum) and *Salmonella* antibody. **B** Illustration depicting the affinity-based sensing chemistry involving antibody-antigen binding for *Salmonella*. **C** Optical images, obtained through Hirox imaging, showcasing the Au-sensor and Au/ZnO-sensor

Following this, the chip was loaded with the protein A molecule at a concentration of 20 µg/mL, along with the antibody solution sourced from Invitrogen at 50 µg/mL.

Electrode modification

In this research study, two types of electrodes were employed: Au electrode and Au/ZnO-based electrode. To modify both electrodes, a crosslinker called 3,3′-dithiobis (sulfosuccinimidyl propionate) (DTSSP) was used at a concentration of 10 mM, along with *Salmonella* antibody at a concentration of 10 µg/mL in a volume of 250 µL PBS (phosphate-buffered saline). The DTSSP was mixed with the antibody and incubated at 4 °C for 30 min. Next, the electrodes were washed with the PBS solution. Following the wash, a mixture of DTSSP and antibody was applied onto the electrodes and incubated for an additional 30 min at room temperature. After this incubation period, 5 µL of the real sample was applied to the electrode and allowed to incubate for 9 min. Subsequently, data was recorded using laboratory-based reference potentiostat. Electrochemical impedance spectroscopy (EIS) with a minor applied 10 mV AC bias was employed to measure the electrochemical response arising from non-faradaic events taking place at the electrode surface. *Salmonella* binding to specific antibodies caused the double-layer charge at the electrode surface to vary. This was used to measure the alteration in the signal.

Statistical analysis

The presented data are illustrated as mean values with the standard error of the mean (SEM). The dataset is derived from three replicates ($N = 3$), and both inter-assay and intra-assay variations fall within the acceptable range of < 10%, adhering to Clinical & Laboratory Standards Institute (CLSI) guidelines. Statistical analyses were carried out using GraphPad Prism version 10.01, developed by GraphPad Software in La Jolla, CA, USA, with additional analyses performed using Origin 2023. Data modeling and equivalent circuit study were performed using Zview software, followed by Bio-render for creating the graphics of the reaction mechanisms.

Result and discussion

Surface characterization of the modified electrodes

In the present study, we applied affinity-based functionalization for *Salmonella* detection. Our first step involved chemically binding the thiol-based crosslinker (DTSSP) to the Au-electrode, which is followed by the subsequent immobilization of the *Salmonella* antibody to the surface of crosslinker DTSSP as shown in Fig. 1A. The functionalization of the sensor was validated using the Fourier transform infrared spectroscopy (FTIR). In this validation process, an Au-film was deposited onto a glass surface using electron vapor deposition. This surface was then modified with the antibody, mirroring the procedure detailed in the "Electrode modification" section for the Au-sensor platform. The FTIR spectra of the synthesized sensor provide evidence for the existence of the antibody along with the Au and Au/ZnO. The strong peak at 1645 cm^{-1} is attributed to the carbonyl stretch of NHS in all three FTIR spectra [24]. Figure 1B provides clear evidence of the C-N–C stretch of the NHS at 1260 cm^{-1} and the N–O-C esters peak at 1043 cm^{-1}. The subsequent disappearance of these peaks at 1260 and 1043 cm^{-1} is indicative of the immobilization of the capture probe. This immobilization is a direct result of the conjugation between the antibody and DTSSP [32, 33]. The optical images captured for the designed Au and Au/ZnO sensor platform are depicted in Fig. 1C.

SPR studies

This study employs surface plasmon resonance (SPR), an optical analytical technique, to comprehensively investigate and quantify surface-protein interactions. The focus is on assessing the binding kinetics of antibodies and antigens, crucial for validating antibody efficacy. By conducting SPR experiments and utilizing a binding-fit model, the dissociation constant (KD), inversely proportional to the affinity constant, is determined. This method proves sensitive in capturing the signal variation during the binding process between the receptor antibody and target analyte, confirming the accuracy of the proposed immunoassay stack [34]. The label-free analysis facilitated by SPR biosensors allows for examining the binding of analyte molecules in biofluids to recognition components on the sensor substrate. This optical biosensing approach precisely quantifies binding kinetics by leveraging variations in the detector's refractive index due to binding interactions, affirming its efficacy in characterizing surface-protein interactions [35, 36].

The binding affinity of the *Salmonella* antibody was assessed by immobilizing it at a concentration of 50 mg/mL onto a gold-carboxyl-functionalized sensor chip. This immobilization was achieved through EDC/NHS cross-linkage with the assistance of protein A. The SPR temporal spectra in Fig. 2 illustrate this process. The addition of HCl for sensor conditioning (represented as 1) results in a small dip. The subsequent injection of the EDC/NHS mixture (step 2) modulates the refractive index, causing a spike in the response due to binding to the carboxyl-ends of the sensor. Protein A injection (step 3) further orients the immobilization surface. In step 4, the blocking buffer injection deactivates unused carboxyl groups, resulting in a bulk change in the optical response. Subsequently, during antibody injection in step 5,

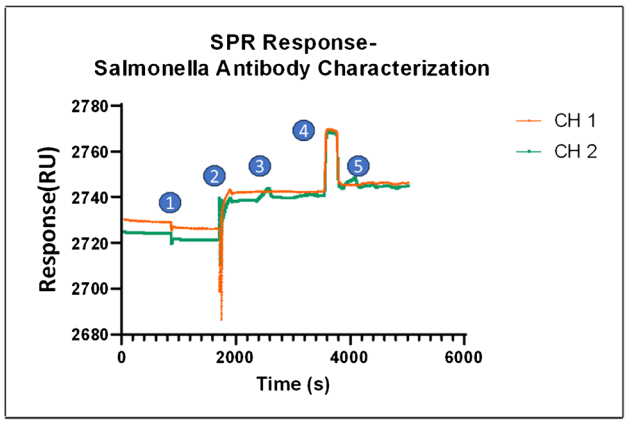

Fig. 2 Utilization of SPR signal for the functionalization and characterization of the *Salmonella* antibody on the sensor

there is a specific increased step change in channel 2 (active channel) compared to channel 1 (control). In binding characterization, a commonly utilized affinity metric relies on the association constant (Ka) and dissociation constant (Kd) to compute the equilibrium dissociation constant (KD). This constant is inversely proportional to the affinity between the ligand and the target analyte.

$$KD = Kd/Ka$$

The binding attributes of antibody interaction with the substrate are illustrated in Supplementary Fig. S1 using the fitted model. The nanomolar sensitivity of binding, where a lower KD (5×10^{-9}) value signifies higher affinity (as explained earlier), is evident. The SPR study plays a crucial role in characterizing antibody binding, thus validating both the electrochemical immunoassay and the sensor stack.

Zeta potential characterization

The surface charge and stability of the *Salmonella* can be characterized through zeta potential measurements. Under an applied electric field, *Salmonella* reacts with co-ions and counterions of the buffer until it reaches a charge-neutral state and advances to the opposite electrode with the help of electrostatic forces [37, 38]. This electrophoretic mobility presents the zeta potential and charge state of *Salmonella*. This study aims to understand the regulation of the electrochemical surface charge within the electric double layer and the stability of *Salmonella* when it is distributed in potable water in varying dilutions.

The *Salmonella* bacteria are encased in a slender peptidoglycan cell wall and wrapped in an outer membrane that includes lipopolysaccharide. At neutral pH, *Salmonella* bacteria have a net negative charge because of the existence of peptidoglycan and lipopolysaccharide which is abundant in carboxyl, amine group, and phosphate group. Figure 3A demonstrates a simplified representation of the potential distribution in the electric double layer (EDL). The double layer consists of two parts, the Stern layer: this is the layer of ions and molecules that strongly adhere to the electrode's surface. The diffuse layer: beyond the Stern layer, there's a diffuse layer of ions that are less tightly bound. This layer contains negatively charged ions, helping to maintain charge neutrality in the solution [39]. The binding interactions between the *Salmonella* antibody and antigen on the electrode surface alter the electrical double layer [39]. Variances in charge during the binding of *Salmonella* antibody (10 µg/mL) with 10^1, 10^2, 10^3, 10^4, 10^5, and 10^6 CFU/mL of *Salmonella* bacteria were determined through ζ potential analysis. The system's sensitivity to target analyte concentrations is evaluated by the sensing platform through electrochemical

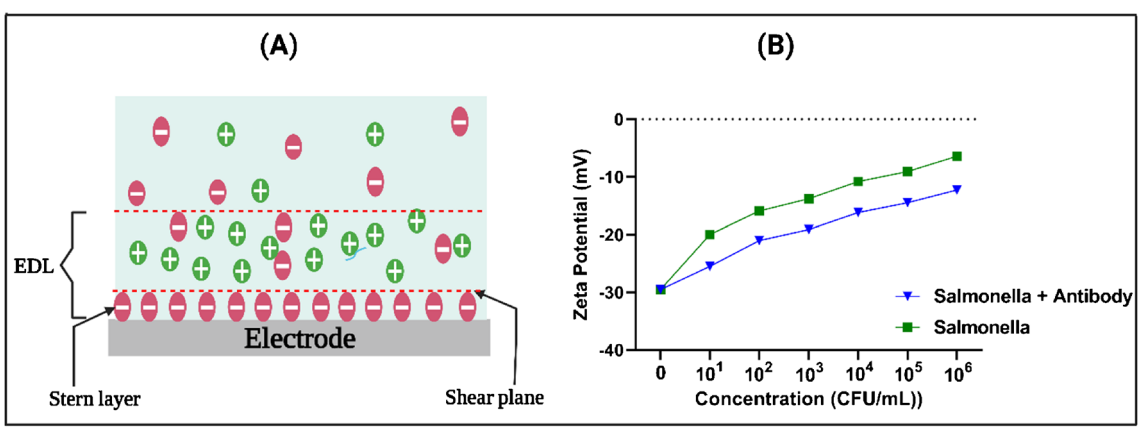

Fig. 3 **A** Schematic representation of interaction between the antigen and the antibody-modified platform induces alterations in the double layer at the electrode–electrolyte interface. **B** Zeta potential of *Salmonella* and *Salmonella* antibody in potable water with various volumetric dilution concentration

detection. Therefore, the binding in the presence of an electrical double layer can be understood through the evaluation of the zeta potential [40]. Figure 3B illustrates the results of the zeta potential study. The introduction of *Salmonella* in potable water changes the ζ potential from − 29.5 ± 0.26 to − 6.43 ± 0.44. In the absence of the *Salmonella* antibody, the ζ potential displayed significant change when the dose concentration increased from 10^1 CFU/mL (a low dose) to 10^6 CFU/mL (a high dose). However, the ζ potential changes from − 29.5 ± 0.26 to − 12.3 ± 0.21 in the presence of the *Salmonella* antibody (10 mg/mL) with the same concentration of *Salmonella* bacteria in potable water shown in Fig. 3B. The introduction of *Salmonella* antibodies had a notable impact on the zeta potential. The presence of the antibody results in a change in the overall ionic concentration of the solution. This change in ionic concentration leads to a shift in the zeta potential from − 6.43 to − 12.3 mV, indicating that the antibodies effectively bind to the *Salmonella* bacteria. These changes in the zeta potential can be associated with modifications in the capacitive behavior, specifically the constant phase element (CPE) parameter, within the parametric fit of the impedance data [41], a detailed discussion is given in the next section.

Investigating the modulation of the electrical double layer (EDL)

Non-faradaic EIS is an effective technique that captures the binding events at the electrode–electrolyte interface over a broad frequency range [42, 43], with minor voltage disruptions. This technique proves valuable in evaluating the performance of immunosensors for *Salmonella* detection in potable water. The study highlights the advantages of utilizing low frequencies for sensing performance metrics, leveraging a high signal-to-noise ratio, and facilitating seamless integration into portable electronics for practical application. This aligns well with our EIS data collection approach, emphasizing the efficacy of non-faradaic techniques in capturing binding events and optimizing sensor performance for *Salmonella* detection in real-world water samples.

The electrode–electrolyte interface is comprised of a bulk diffused layer and an EDL which can be represented as an equivalent circuit to emulate the binding mechanism. The EDL has two parts: a dense stringent layer formed by ions that are attracted to the electrode surface due to the opposite charges and a diffuse layer [44]. When mirrored as an equivalent circuit, the changes in impedance during the binding of molecules can be accredited to changes in the resistance of charge transfer or the capacitance of the double layer, based on the buffer solution that was utilized. In this work, we contrast the efficacy and sensing performance of Au and Au/ZnO sensors for detecting *Salmonella* in potable water across the various concentrations, including control (ZD), 10^1, 10^2, 10^3, 10^4, 10^5, and 10^6 CFU/mL doses. In Fig. 4, the curves of calibration dose responses, or CDR, in detecting *Salmonella* were plotted as impedance change from the baseline without any target molecule and then fitted to Randles' equivalent circuit model for further analysis. Figure 4A, B represents Randles equivalent circuit model

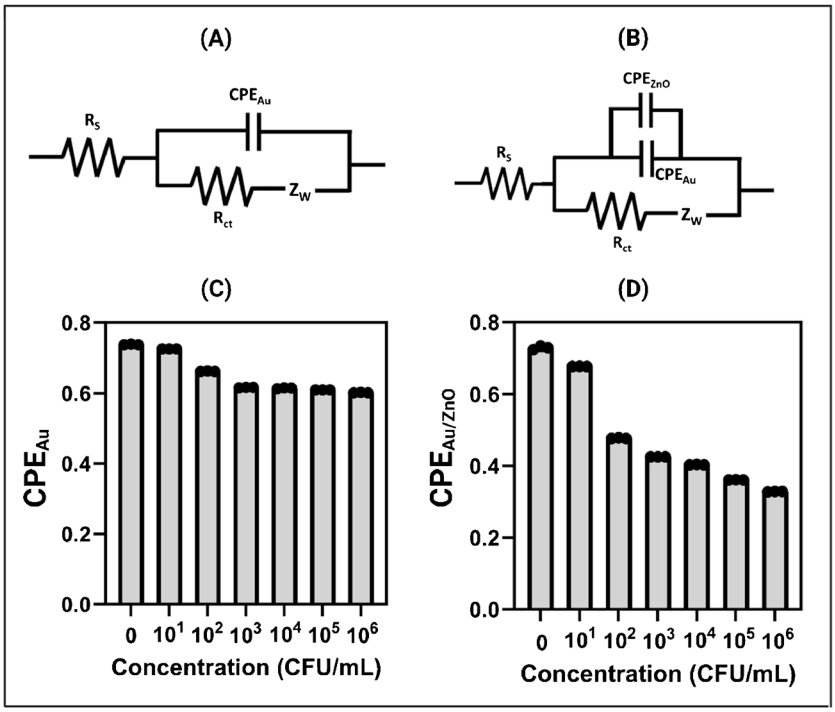

Fig. 4 **A**, **B** The equivalent circuits for the corresponding Au and Au/ZnO-sensor platform. **C**, **D** Represents electrical double-layer capacitance variation with different concentrations of *Salmonella* spiked in potable water

for the Au-sensor and Au/ZnO-sensor. The solution resistance (R_S) represents the bulk resistance in parallel RC circuits at lower frequencies, along with the charge-transfer resistance (R_{ct}) and constant phase element (CPE_{Au}, $CPE_{Au} + CPE_{ZnO}$). There is no significant change in Rct when concentration changes shown in Supplementary Fig. S2. However, the constant phase element (CPE) that exhibits maximum changes in the charged double layer (CPE_{Au}) for Au-sensor was observed throughout the concentration (ZD to 10^6 CFU/mL) from 0.73 to 0.6 µF (as shown in Fig. 4C), while a decrease from 0.7 to 0.3 µF was seen in the $CPE_{Au/ZnO}$ value of the Au/ZnO sensor depicted in Fig. 4D. This change is attributed due to the modulation of the double layer charge at the electrode surface due to the binding of the target antigen to specific antibodies [45]. The lowest frequency yielded the highest signal-to-noise ratio, prompting its selection for subsequent analysis and the creation of the calibrated dose–response plot. The contrast in charge double layer values demonstrates the changes in the interfacial behavior and electrochemical properties of the sensors due to the variation of *Salmonella* concentrations. Furthermore, the enhanced accuracy and sensitivity observed in the ZnO-modified gold electrode compared to the unmodified gold electrode can be attributed to the unique properties of ZnO nanomaterials. ZnO, with its high surface area and favorable electrochemical characteristics, provides an improved platform for the immobilization of DTSSP and *Salmonella* antibodies. The ZnO nanomaterial not only improves the binding of biomolecules more effectively but also accelerates charge transfer kinetics during electrochemical impedance spectroscopy (EIS) measurements. This assertion is supported by evidence in the form of observed changes in impedance measurements, demonstrating enhanced charge transfer efficiency associated with the presence of ZnO nanomaterials [46]. The increased surface roughness and active sites on the ZnO-modified electrode contribute to a higher surface area available for *Salmonella* binding, leading to an amplified signal response. Moreover, ZnO's inherent semiconductor properties may influence the charge transport and electron transfer kinetics, contributing to the heightened sensitivity observed in the detection of *Salmonella*. Overall, the synergistic effects of ZnO modification on the gold electrode result in a sensor with superior accuracy and sensitivity, making it a promising platform for *Salmonella* detection applications.

Spike and recovery study

A spike and recovery study is a quality control method used to assess the accuracy of an analytical procedure. It involves adding a known amount of the target analyte (spike) to a sample, measuring its recovery, and comparing it to the expected value. This helps verify the reliability of the analytical method.

In the spike and recovery experiments, *Salmonella* was randomly spiked at concentrations ranging from control/ZD to 5, 50, 500, and 5000 CFU/mL on the Au and Au/ZnO sensors. The study involved $N = 8$ replicates. A Nyquist plot, derived from electrochemical impedance spectroscopy (EIS), visually represents impedance data by showcasing the real part (Zreal) on the *x*-axis and the imaginary part (Zimg) on the *y*-axis. Recognized for its characteristic semicircular nature, the Nyquist plot offers insightful details about the electrochemical processes within a system shown in Supplementary Fig. S3. Simultaneously, the calibration dose response (CDR) serves as a crucial method for assessing a sensor or analytical system's performance in relation to varying concentrations of a target analyte. In the EIS context, CDR involves measuring the impedance response across a spectrum of analyte concentrations, and the resulting dataset constructs a calibration curve, elucidating the correlation between impedance changes and analyte concentration. The CDR plots obtained for both sensors demonstrated a response across a wide concentration range, as illustrated in Supplementary Fig. S4A, B. The Au sensor (depicted in Fig. 5A) manifested a strong correlation with an R^2 value of 0.96, and the percentage error consistently stayed below 20% within the given concentration range. This implies that the fabricated sensor is closely aligned with the actual *Salmonella* concentrations present in the potable water. Furthermore, the LoD (calculated based on the signal-to-noise (S/N) threshold such as specific signal threshold (SST) = baseline signal + 19.6 × standard deviation of baseline) for the Au sensor was determined to be 0.9 CFU/mL, indicating its ability detect low levels of *Salmonella* contamination. The Au/ZnO-based sensor (Fig. 5D) exhibited a correlation $R^2 = 0.97$ and a percentage error of < 15%. This indicates a reliable match between the estimated and actual *Salmonella* concentrations detected by the sensor. Moreover, the LoD for the Au/ZnO sensor was determined to be 0.6 CFU/mL, signifying its capacity to detect *Salmonella* even at low concentration. Au and Au/Zno-LoD are comparable but analytical resolution and quantification capability are significantly greater in Au/ZnO modified sensor. Figure 5B, E shows the spike concentration plotted against the percentage of recovered *Salmonella* concentration (CFU/mL) in the Au and Au/ZnO sensors, respectively. The recovery percentage fell within an error range of ± 10–15%. The obtained results were graphically represented, and clear distinctions were observed in the corresponding figure (Fig. 5C, F). It illustrates the division of the plot into separate sections based on the concentrations (low, mid, and high) and the ZD (control group). This visual representation provides a concise overview of the experimental findings, enabling easy interpretation and analysis.

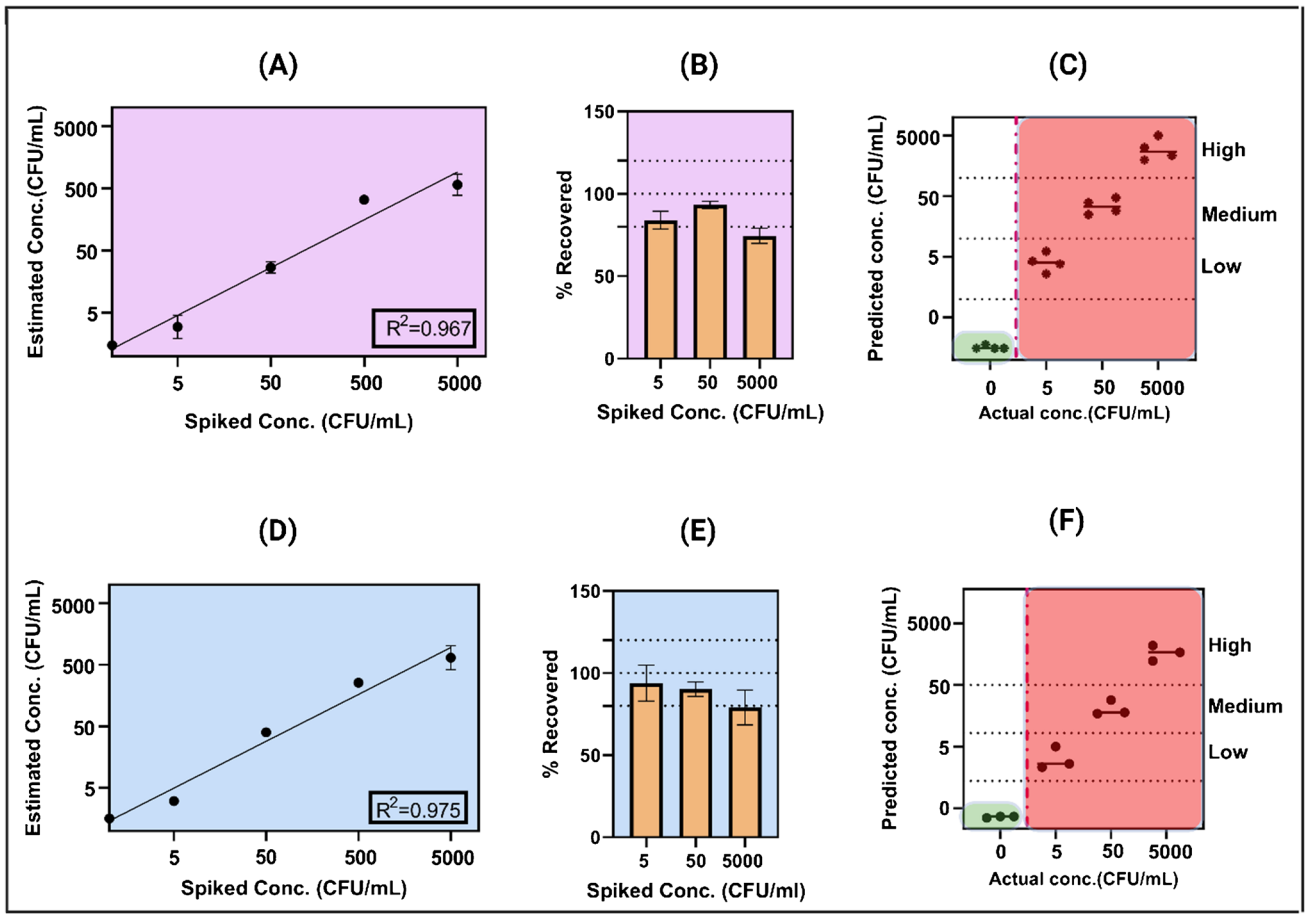

Fig. 5 **A D** Represents spike and estimated signal response on Au and Au/ZnO sensors, respectively with spiked dose concentrations control (ZD), 5 to 5000 CFU/mL in potable water. **B**, **E** The recovery graph shows the system's performance in detecting *Salmonella* at different concentrations. **C**, **F** Performance of the sensor at low, medium, and high concentration

Cross-reactivity and correlation study

A sensor platform engineered for real-time applications is of utmost importance for its selective performance. Cross-reactivity was performed on the modified sensor platform by spiked cross-reactive antigens at varying concentrations alongside analytes of interest in potable water shown in Supplementary Fig. S5. The response plot for the cross-reactive study shows a 16% change in total impedance after high doses of *E. coli* on the *Salmonella*-Ab sensor platform and less than 10% change after medium and low doses. These results indicate that the sensor platform maintains a high level of specificity, as reflected by the minimal cross-reactivity observed at low, medium, and high concentrations of structurally similar pathogens.

After a successful performance study of Au and Au/ZnO platforms to detect *Salmonella* in potable water, a test was conducted to assess the feasibility of translating to a portable device. A gold-modified zinc oxide platform compares with the portable and benchtop devices. In order to confirm the viability and suitability of deploying the sensor in the field, the Pearson correlation assessment was conducted between the portable device and the laboratory potentiostat reference method. This validation process involved executing a parametric correlation test, utilizing linear alignment between the portable device and the laboratory instrument results. The portable device demonstrated an excellent correlation with the data obtained from the reference instrument, as evidenced by a Pearson correlation coefficient (Pearson *r*) value of 0.9872, as illustrated in Fig. 6A.

Furthermore, a Bland–Altman (BA) analysis was conducted to assess the correlation level and examine potential biases in the data across varying dose ranges between the portable device and the reference method. The BA analysis provides insights into any systematic biases or offsets in measurements between the two methods. In Fig. 6B, the *x*-axis signifies the average of the concentration doses, and the *y*-axis depicts the concentration difference, representing the delta changes between the two methods for each sampled measurement. Compared to the upper range of the mean

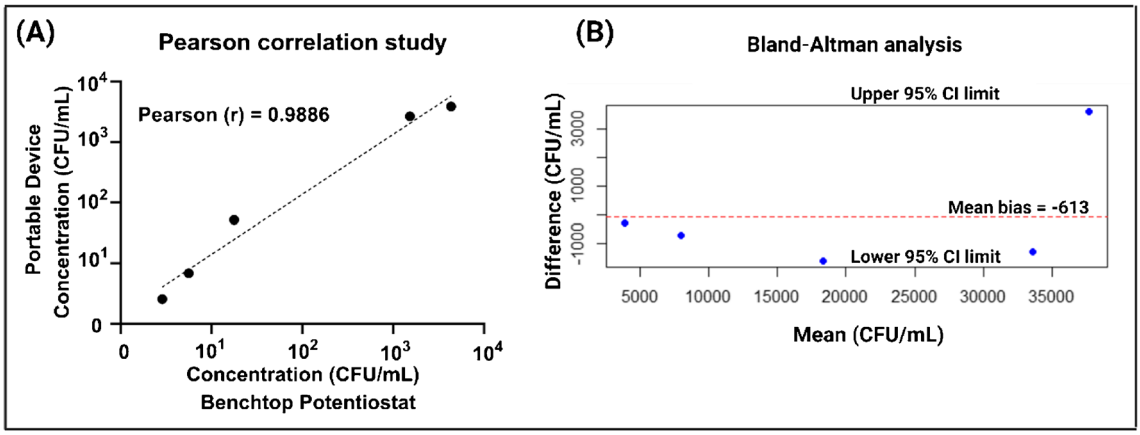

Fig. 6 **A** Investigating the parametric correlation between a portable device and a benchtop reference device for five concentrations reveals a linear fitting. **B** Bland–Altman plots to evaluate the level of correlation showing a mean bias of − 613 with a 95% confidence limit

concentration, the system exhibits a mean bias of − 613, indicating a statistically insignificant difference in measurement cycles between the two devices. Furthermore, every test sample is within ± 1.96 times the standard deviation, constituting the upper and lower 95% confidence interval, thereby confirming a commendable agreement between both methods. Moreover, the averages of the random distributions of points situated above and below the mean bias line imply the lack of over/under compensation or bias. This ensures consistent and unbiased predictions of adulteration levels by either instrument.

Reproducibility and repeatability

Au and Au/ZnO sensors have the potential in capturing low concentrations of *Salmonella* in potable water as shown in the detection of *Salmonella* presented above. Inter-assay reproducibility and intra-assay repeatability of the system

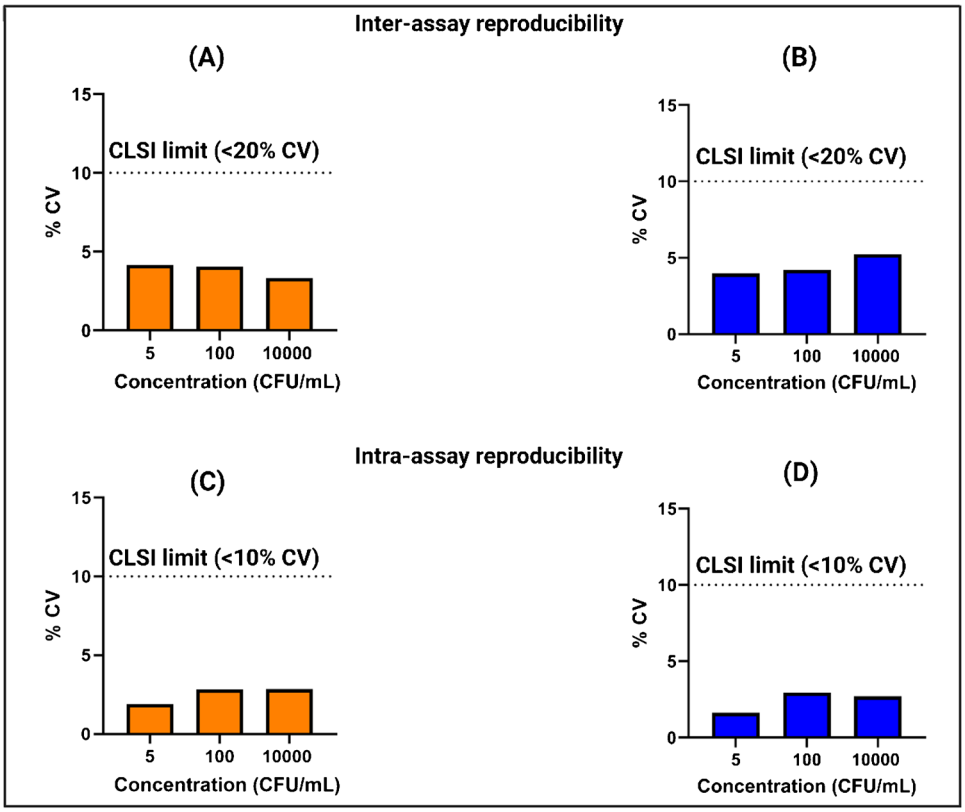

Fig. 7 Reproducibility and repeatability tests with low, medium, and high dose responses, showing statistically consistent performance, confirmed by % CV. **A**, **B** inter-assay variation in Au and Au/ZnO sensor platform, $N = 5$. **C**, **D** intra-assay variation in Au and Au/ZnO sensor platform, $n = 3$

were assessed in potable water on both sensor platforms to comprehend the response of the system and is shown in Fig. 7A–D, respectively. Figure 7A, B shows the coefficient of variation (% C) for different concentrations of *Salmonella* (low, medium, and high) from $N = 8$ sensors for the Au-sensor and Au/ZnO sensor, respectively. The sensors show a coefficient of variation (CV) < 20% for the different *Salmonella* concentrations which highlights that the inter-assay variability is within the Clinical and Laboratory Standards Institute guidelines [47]. The intra-assay variation of Au and ZnO sensors, as shown in Fig. 7C, D, respectively, was conducted over $n = 3$ cycles and stayed under 10% with different concentrations of Salmonella which is compliant with the CLSI guidelines. The results presented emphasize that the performance for both sensors in detecting *Salmonella* in water matrices is reproducible and repeatable.

Conclusion

The feasibility and effectiveness of utilizing electrochemical impedance spectroscopy (EIS) for rapid and valid detection of *Salmonella* in portable water was demonstrated throughout this paper. The immunoassay-sensing platform, which was designed by immobilizing antibodies on Au and Au/ZnO sensor platforms, exhibited specific and selective recognition of the target pathogen. The proposed system provides rapid detection, achieving a response time of less than 9 min, while utilizing a minimal sample volume of 5 mL for *Salmonella*-contamination testing. The limit of detection (LoD) was calculated at 0.9 and 0.6 CFU/mL for the Au and Au/ZnO sensors, respectively. The successful development of this electrochemical immunoassay-based sensing system prevents promising prospects for the field of bacterial detection in water samples. With its portability and ease of use, the proposed approach holds great potential for on-site and real-time monitoring, facilitating early detection of *Salmonella* contamination in potable water. This research grants knowledge to the growing field of biosensor technology and its application in food safety and public health. The findings pave the way for further advancements in electrochemical-based sensing platforms, offering valuable contributions to the mitigation of bacterial outbreaks and the assurance of safe potable water. By addressing critical challenges in bacterial detection, this study marks a significant step towards enhancing the overall quality and safety of water resources, thereby safeguarding public health and wellbeing.

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Declarations

Competing interests The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Drs. Shalini Prasad and Sriram Muthukumar have a significant interest in Enlisense LLC, a company that may have a commercial interest in the results of this research and technology. The potential individual conflict of interest has been reviewed and managed by The University of Texas at Dallas and played no role in the study design; in the collection, analysis, and interpretation of data; in the writing of the report, or in the decision to submit the report for publication. Portable device and technology platform is a proprietary of EnLiSense LLC.

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